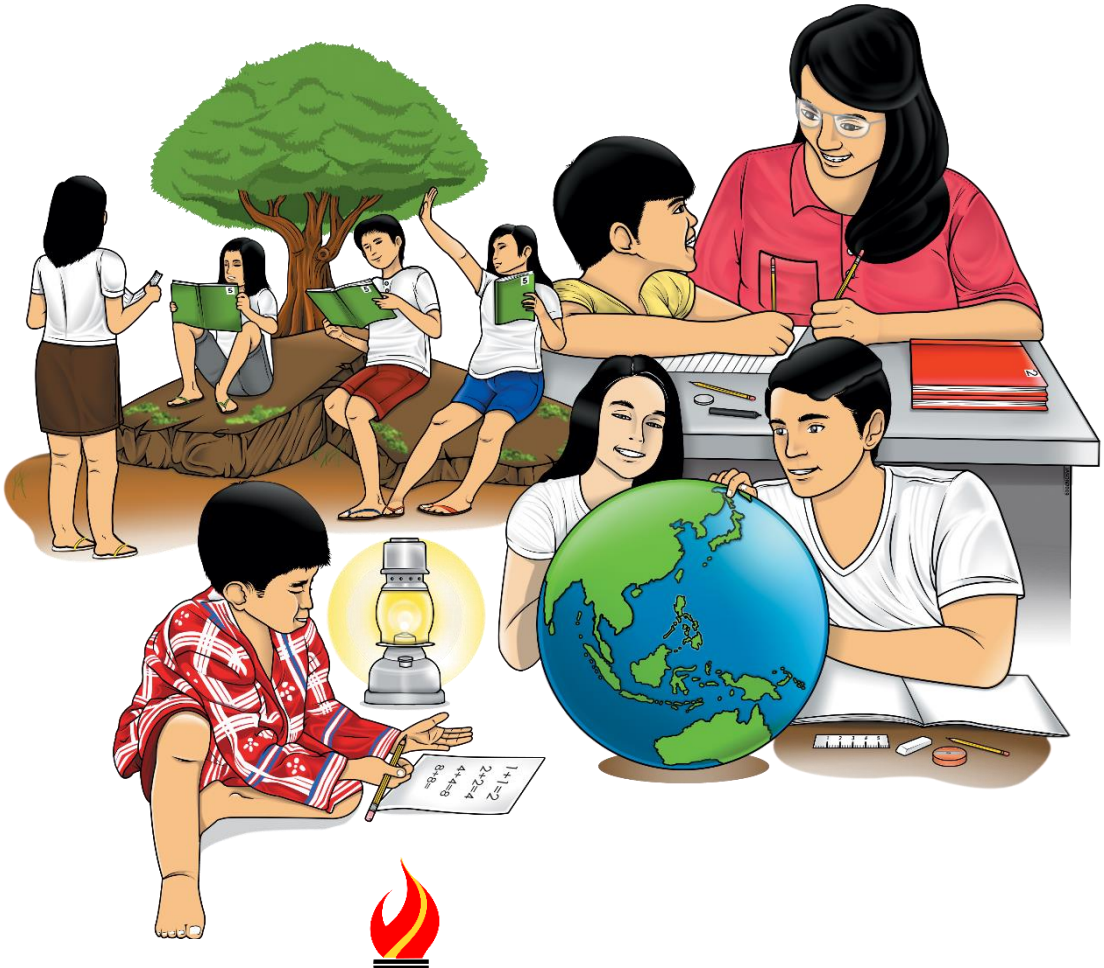
Senior High School

DepED
DEPARTMENT OF EDUCATION

General Physics 1

Quarter 2 – Module 2

Rotational Kinematics, Law of Gravitation, Waves, Sound, Fluid Mechanics, Thermodynamics and Ideal Gases



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General Physics 1
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Quarter 2: Rotational Kinematics, Law of Gravitation, Waves, Sound, Fluid Mechanics, Thermodynamics and Ideal Gases

First Edition, 2020

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Development Team of the Module

Writers: Armando A. Camaña Jr., Jenievie E. Datuin, Camille L. Espino, Lerrie P. Munsod, Amiel L. Vijandre, Felines M. Vijandre

Editor: Maria Lea Q. Prondo, Ed.D. - EPS

Content Validators: Loreta L. Salvador, Ph.D.
Irene C. Cular

Language Editor: Maria Rhea M. Pelayo

Layout Artist: Felines M. Vijandre

Management Team: Dr. Meliton P. Zurbano – OIC-SDS
Filmore R. Caballero – CID Chief
Jean A. Tropel – Division EPS In Charge of LRMS
Maria Lea Q. Prondo, Ed.D. – Education Program Supervisor

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Department of Education – NCR – VALENZUELA CITY

Office Address: Pio Valenzuela St.Marulas, Valenzuela City
Telefax: (02) 292-3247
E-mail Address: sdovalenzuela@deped.gov.ph

What I Need to Know

After going through this module, you are expected to:

1. **Calculate the moment of inertia about a given axis of single-object and multiple-object systems [STEM_GP12REDIIa-1].**

What I Know

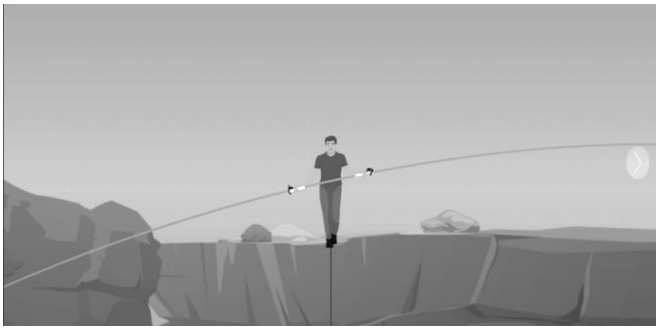
Directions: Fill in the K-W-H-L Chart about rotational inertia on a paper.

K	W	H	L
What do I know?	What do I want to find out?	How can I found out what I learn?	What did I learn?

Lesson 1 Rotational Inertia

What's New

DIRECTIONS: Observe the picture below and answer the guide questions.
Daredevils risk their lives walking along cables at dizzying heights.



Guide Questions:

1. What might cause them to fall?
2. How can they prevent falling?
3. Why does a tightrope walker require a balancing pole?

What is It

Rotational inertia, also called moment of inertia, tells you how difficult it is to rotate an object. It is a measure of how resistant an object is to changes in its rotational motion, and is equal to the product of mass of the object **(m)** with the square of its distance from the axis of the rotation **(r).** In equation, $\mathbf{I=mr^2}$. Consequently, it has SI unit of $\mathbf{kgm^2}$. For point mass object, like the ring, rotational inertia is equal to $\mathbf{mr^2}$**.** However, other objects have different rotational inertia (refer to Table 1). Mechanical systems are often made of many masses that are connected together, or in complex shape. To calculate the total rotational inertia of a system about any axis, simply get the sum of the rotational inertial of each mass, $\mathbf{I=\sum mr^2}$.

For example, the moment of inertia of a pole with a mass at its end is the sum of the individual moment of inertias of each object involved.

Table 1: Moment of Inertia of Various Objects

Object	Moment Inertia
Hollow cylinder (hoop or ring) of radius r	$I=mr^2$
Solid cylinder of disk of radius r	$I=\frac{1}{2}mr^2$
Hollow sphere of radius r	$I=\frac{2}{3}mr^2$
Solid sphere of radius r	$I=\frac{2}{5}mr^2$
Thin rod of length r rotating about its middle	$I=\frac{1}{12}mr^2$
Thin rod or plank of length r rotating about one end	$I=\frac{1}{3}mr^2$

What's More

Activity: **Do the Math on Rotational Inertia**

Material: moment of inertia table (Table 1)

Procedures:

1. Use the moment of inertia table to answer the following problems. Write your solutions on a sheet of paper.
   a. Calculate the moment of inertia of the Earth about its spin axis.
   b. Calculate the moment of inertia of the Earth as it revolves around the Sun.
   c. Calculate the moment of inertia of a hula hoop with mass 2kg and radius 0.5m.
   d. Calculate the moment of inertia of a rod 0.75m in length and mass 1.5 kg rotating about one end.
   e. Repeat step *d* but calculate the moment of inertia about the center of the rod.

Guide Questions:

1. What have you observed with the moment of inertia of objects in **a** to **e**?
2. Why does the moment of inertia vary from one object to another?

What I Can Do

Directions: Answer the question on a sheet of paper.

Two spheres of the same mass are spinning in your garage. The first is 12 cm in diameter and made of solid plastic. The second is 10 cm in diameter but is a thin sphere shell of iron filled with air. Which is harder to slow down? Why?

Assessment

Directions: Write YES if the statement is true and write NO if false.

_____1. The moment of inertia of a body independent of its angular velocity.

_____2. To increase the moment of inertia of a body about an axis, you must place part of the body farther from the axis.

_____3. The moment of inertia of a non-uniform semi-circular wire having mass **m** and radius **r** about a line perpendicular to the plane of the wire through the center is mr^2.

_____4. If a solid ball and a cylinder roll down an inclined plane, the ball reaches the bottom first.

_____5. A ring has greater moment of inertia than a circular disc of same mass and radius because its entire mass is concentrated at the rim in maximum distance from the axis.

What I Need to Know

After going through this module, you are expected to:

1. **Calculate magnitude and direction of torque using the definition of torque as a cross product. [STEM_GP12REDIIa-3]**
2. **Describe rotational quantities using vectors. [STEM_GP12REDIIa-4]**
3. **Determine whether a system is in static equilibrium or not. [STEM_GP12REDIIa-5]**
4. **Apply the rotational kinematic relations for systems with constant angular accelerations. [STEM_GP12REDIIa-6]**
5. **Solve static equilibrium problems in contexts but not limited to see-saws, cable-hinge-strut system, leaning ladders, and weighing a heavy suitcase using a small bathroom scale. [STEM_GP12REDIIa-8]**
6. **Determine angular momentum of different systems. [STEM_GP12REDIIa-9]**
7. **Apply the torque-angular momentum relation. [STEM_GP12REDIIa-10]**

What I Know

Directions: Identify the term being described in each statement. Choose from the choices below and write your answer on a sheet of paper.

1. It is the amount of force needed to twist an object.
2. It refers to the rate of change of angular position of a rotating body.
3. The time rate of change of the angular velocity is called _____.
4. It is quantified by measuring how far the body is rotated from the reference position and can be measured in degrees, radians or revolutions.
5. _____ is the product of angular velocity of the body or the system and its moment of inertia with respect to the rotation axis, and is directed along the rotation axis.

Angular acceleration	Angular momentum	Angular position	Angular velocity	Torque

Lesson 2 Rotational Kinematics and Rotational Dynamics

What's New

DIRECTIONS: Put a check (✓) on the space before each the item which illustrates rotational motion and cross (X) if not.

__1.

__2.

__3.

__4.

__5.

What is It

Rotational Kinematics

The kinematics of rotational motion describes the relationships between angular position (**θ**), angular velocity (**ω**), angular acceleration (**α**), and time (**t**). To derive the equations for rotational motion, recall the kinematics for translational motion. Table 2 shows how kinematics for rotational motion is completely analogous to translational motion. Table 3 shows how angular variables and linear variables are related. These equations can be used to solve rotational and linear kinematics problem in which α *and* $\boldsymbol{a}$ *are constant.*

Table 2: Rotational Kinematics Equations

Rotational Motion	Translational Motion
$\theta=\omega t$	$x=vt$
$\omega_f = \omega_i + \alpha t$	$v_f = v_i + at$
$\theta=\omega_i t+{}^1/_2\alpha t^2$	$x=v_i t+{}^1/_2 a t^2$
$\omega_f^2 = \omega_i^2 + 2\alpha\vartheta$	$v_f^2 = v_i^2 + 2ax$

Table 3: Rotational and Linear Variables

Rotational	Linear	Relationship
θ	x	$\theta = \frac{x}{r}$
ω	v	$\omega = \frac{v}{r}$
α	a	$\alpha = \frac{a}{r}$

Sample Problem: A cylinder of radius 20 cm is rolling along the surface with constant speed of 80 cm/s.

a. What is the rotational speed of the cylinder about its axis?

b. What is the magnitude and direction of the acceleration of the points on its surface?

Solutions:

a. Rotational speed $\omega = v/r = 80/20 =$ **4rad/s**

b. Since there is no acceleration for cylinder, $\alpha=0$ and we have just radial acceleration

$a_r = \omega^2 r =$ **3.2m/s²**

Rotational Dynamics

Have you ever pushed a merry-go-round? In order to change its angular velocity, you know that force is needed to do it. We have discussed the angular version of distance, velocity and acceleration. Now, let us talk about the angular version of force.

The angular version of force is torque **τ**. It is also referred to as the moment of force, rotational force, or tuning effect. It is defined as the cross product between the rotational force $\vec{\mathbf{F}}$ being applied and the radial distance $\vec{\mathbf{r}}$ from the axis of rotation. In equation, $\vec{\boldsymbol{\tau}}=\vec{\mathbf{F}}\times\vec{\mathbf{r}} = \mathbf{r}\perp\mathbf{F}$. We use **θ** to represent the angle between the force and the object, $\boldsymbol{\tau}=\mathbf{rF\sin\theta}$.

Static equilibrium is a form of equilibrium that occurs when an object is at rest. There are two conditions with static equilibrium; the translational acceleration should be zero as well as the rotational acceleration. In other words, an object should have translational ($\sum \boldsymbol{F} = \mathbf{0}$) and rotational equilibrium ($\sum \boldsymbol{\tau} = \mathbf{0}$).

Sample Problem: Consider the man pushing the playground merry-go-round. He exerts a force of 250 N at the edge of the merry-go-round and perpendicular to the

radius, which 1.50 m. How much torque does he produce? Assume that friction on the merry-go-round is negligible.
Given: r = 1.50 m F = 250 N $\theta=90^0$
Solutions: $\tau=rF\sin\theta$ = (1.50m)(250N)sin(90) = **375N·m**

Torque-Angular Momentum Relations

Have you asked how an ice skater manage to spin faster and faster simply by pulling her arms in? Why does the ice skater not have to exert a torque to spin faster? The answer is a new conserved quantity called angular momentum.

Angular momentum **L** is the rotational equivalent to linear momentum. Both concepts deal with how quickly something is moving and how difficult it is to change that speed. For single particle, it is the cross product of position **r** and linear momentum **p.**

$$\mathbf{L}=\vec{\mathbf{r}}\times\vec{\mathbf{p}} = \mathbf{rp}\sin\boldsymbol{\theta}$$

where θ is the angle between r and p. Angular momentum is expressed in kgm/s^2. For rigid body, it is the product of an object's moment of inertia **I** and its angular velocity **ω**, **L= Iω .** The net torque can be calculated by taking the time derivative of angular momentum; thus, we have $\frac{d\mathbf{l}}{dt} = \sum\boldsymbol{\tau}$.

What's More

Activity: **Rotational Kinematics and Dynamics**
Procedure: Solve the following word problems in a sheet of paper.

1. A deep-sea fisherman hooks a big fish that swims away from the boat pulling the fishing line from his fishing reel. The whole system is initially at rest and the fishing line winds from the reel at a radius of 4.50 cm from its axis of rotation. The reel is given an angular acceleration of 110 rad/s2 for 2.0 s.
   1. What is the final angular velocity of the reel?
   2. At what speed is the fishing line leaving the reel after 2.0 sec elapses?
2. A wooden plank is balanced on a pivot, as shown below. Weights are placed at various places on the plank. Consider the torque on the plank caused by weight A.
   a. What force, precisely, is responsible for this torque?
   b. What is the magnitude (value) of this force, in Newtons?
   c. What is the moment arm of the torque produced by weight A?
   d. What is the magnitude of this torque, in N·m?
   e. Repeat parts (a – d) for weights B and C.
   f. Calculate the net torque. Is the plank balanced? Explain.

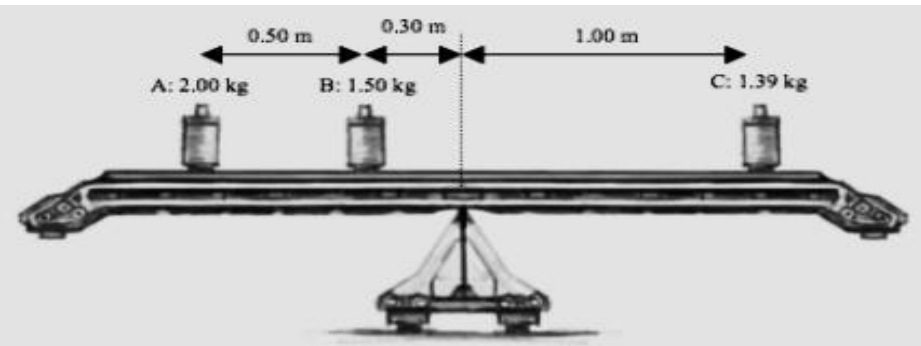


Figure 1. A wooden plank on a pivot

What I Can Do

Directions: Solve the problem. Write your answer on a sheet of paper.

You spin up some pizza dough from rest with an angular acceleration of 5rad/s^2.

a) How many radians has the pizza dough spun through in the first 10 seconds?
b) How many times has the pizza dough spun around in this time?
c) What is its angular velocity after 5 seconds?

Assessment

Directions: Read the following statements. Write True if the statement is correct and False if otherwise.

____1. Applying a longer lever arm to an object so it will rotate produces more torque.
____2. When two children of different weights balance on a seesaw, they each produce equal torques in opposite directions.
____3. A person trying to fix the car with a wrench must turn a bolt counter clockwise to loosen it. The direction of the torque is away from the page and is positive.
____4. When the intrepid mechanic uses a 6-inch-long wrench, he manages to exert a torque of 35Nm. A neighbor loans him a 24-inch-long wrench. If he exerts the same force on the longer wrench as on the shorter one, the maximum torque the original mechanic could provide with the longer wrench is 140 Nm.
____5. One side of a seesaw carries a 21 kg mass four meters from the fulcrum and a 25.5 kg mass two meters from the fulcrum. To balance the seesaw, a 15 kg mass should be placed nine meters from the fulcrum on the side opposite the first two masses.

What I Need to Know

After going through this module, you are expected to:

Use newton's law of gravitation to infer gravitational force, weight, and acceleration due to gravity. [STEM_GP12G-IIb-16]

What I Know

DIRECTIONS: Choose and underline the appropriate word to complete the statement.

Newton's law of universal (gravitation, planetary) states that the force of gravity affects everything with mass in the universe. It also states that the strength of (energy, force) between two bodies depends on their (volume, mass) and the (distance, displacement) between them. Objects with (greater, smaller) mass have a stronger force of gravity between them. Objects that are closer together have a (stronger, weaker) force of gravity between them.

Lesson 3 Newton's Law of Universal Gravitation

What's New

Directions: Read the short story board below and answer the questions.

Figure 2. Story board

Questions:

1. Who was the physicist the two girls are talking about?
2. Can you describe what phenomenon this physicist became famous for? If so, what do you call this phenomenon?

What is It

What keeps the planets in our universe to stay in orbit? Why can't the moon travel like a comet? During the 17th century, Sir Isaac Newton proposed that there must be an inward force that acts on the planets as to why they travel along their orbit. He proposed to call this force *gravity.*

Newton took his idea a step further with his formulation of the **Law of Universal Gravitation** which describe the mutual attraction between two objects. The Law of Universal Gravitation states that this force is directly proportional to the object's mass but inversely proportional to the square of the distance between the objects. In equation form, this law is expressed as:

$$F_g = \frac{Gm_1m_2}{r^2}$$

where:

F_g = Force of Gravity

G = gravitational constant = 6.67×10^{-11} N·m²/kg²

r = the distance between the objects center of mass

m_1 = mass of object 1

m_2= mass of object 2

Sample Problem

Yuri the astronaut was on the moon and wanted to determine its acceleration due to gravity of the moon. From where he was standing, Yuri was 1737.1 km from the moon's core.

Given: m_y = 97 kg m_m = 7.35×10^{22} kg r = 1737.1 km or 1.73×10^6 m

Solution:

Remember Newton had another equation about force, F=ma. If we use the F = mg to measure the gravitational force on Yuri, we can equate this to the law of universal gravitation which also measures the gravitational force on Yuri.

$$m_y g = \frac{G m_y m_m}{r^2}$$, Cancel the m_y on both sides.

$$g = \frac{G m_2}{r^2} = \frac{6.67x10^{-11}\frac{Nm^2}{kg^2}(7.35x\,10^{22}\,kg)}{(1.73\,x10^6\,m)^2} = \mathbf{1.64\ m/s^2}$$

What's More

Directions: Show your complete solution on a separate sheet of paper.

Two satellites of equal mass are put into orbit 25 m apart. The gravitational force between them is 2.0 x10^{-7} N. Find the mass of each satellite.

What I Can Do

Directions: Give an insight about the scenario and defend your answer.

Dave and Ava were arguing about the validity of Newton's Law of Universal gravitation. Dave was against this law and was pointing out that he didn't feel any mutual attraction toward Ava or even towards any object on earth. At a loss of words, Ava just kept staring at the solar system, looking for answers to argue with Dave. Could you defend Ava?

Assessment

Directions: Read the problem below and show your complete solution.

Mr. Spock was floating at distance of 2.25 x10^8 m from earth, suddenly an unidentified spacecraft warps at a distance of 4000 km from him. Unable to ask for any assistance, would an 82-kg Mr. Spock gravitate towards earth or the unidentified spacecraft? The spacecraft was analyzed to be of m = 9.30 x10^{22} kg.

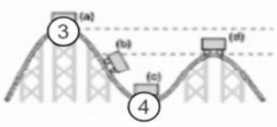
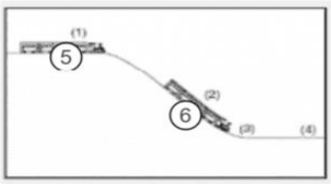
What I Need to Know

After going through this module, you are expected to:

1. **Discuss the physical significance of gravitational field. [STEM_GP12Red-IIb- 18]**
2. **Apply the concept of gravitational potential energy in physics problems. [STEM_GP12Red-IIb- 19]**

What I Know

Directions: Tell whether the following scenarios show minimum or maximum Gravitational Potential Energy. Write minimum or maximum for your answer.

Scenario 1 Gravitational Potential Energy	Scenario 2 Gravitational Potential Energy	Scenario 3 Gravitational Potential Energy
		

1. _____ 4. _____
2. _____ 5. _____
3. _____ 6. _____

Lesson 4 Gravitational Potential Energy

What's New

In using the hammer to perform work on the nail, why do we raise the height of the hammer? Is it to gain more speed or momentum? What you don't realize, to perform more work on the nail and to drill that nail harder to the board, would require more *gravitational potential energy. By elevating the height of the hammer, you have more potential to do work on the nail.*

What is It

Which force is similar to gravity? If you thought of *electromagnetic force*, you are correct! We know any objects from outer space that comes into range of the gravitational field of earth will be influenced towards its crust, just the same way iron fillings fly toward a magnet if it comes close enough.

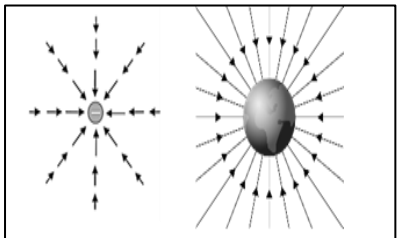


Figure 3. Comparison of Field lines

Observing the gravitational field and electric field lines shown in Fig. 3, the *electric field* lines of a negatively charge particle act similarly as the *gravitational field* lines. Strength of the gravitational field farther from earth will decrease as the distance between each field lines continue to increase. Gravitational Field Strength is numerical equivalent to the acceleration due to gravity and can be expressed in equation as: $\boldsymbol{g = G\frac{M}{r^2}}$

where:

g= Gravitational Field strength G = gravitational constant

r= radius

GRAVITATIONAL POTENTIAL ENERGY

Gravitational potential energy (U_g) is the energy stored in an object as the result of its vertical position or height. It is expressed in equation as: $\mathbf{U_g = mgy}$

where:

m = mass of the object

g = acceleration due to gravity

y = height

Sample Problem:

A box has a mass of 5.8 kg. The box is lifted from the garage floor and placed on a shelf. If the box gains 200 J of Potential Energy, how high is the shelf?

Given: m_b = 5.8 kg U_g = 200 J

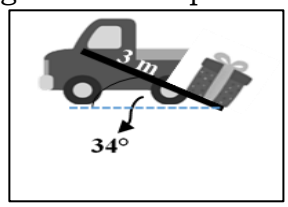
Solution

Derive an equation for y: $\boxed{y = \frac{U_g}{mg} = \frac{200\ J}{(5.8\ kg)\left(9.8\frac{m}{s^2}\right)} = \mathbf{3.52\ m}}$

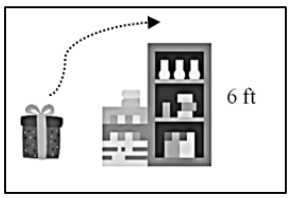
What's More

Directions: Inspect each situation below then determine the gravitational potential energy. Show you answer with complete solution on the box provided.

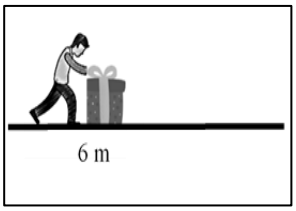
1. Which among the cases places the 5-kg gift in a position with the highest gravitational potential energy? _____



Case A



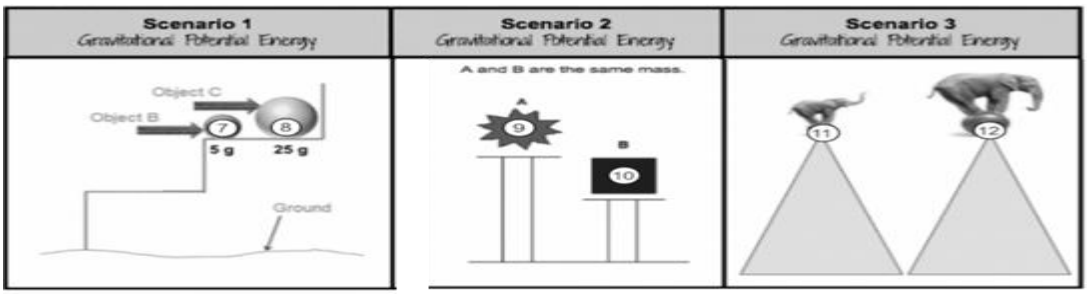
Case B



Case C

What I Can Do

Directions: Write MIN if the scenario shows minimum GPE and MAX if it shows maximum GPE.



7. _____
8. _____
9. _____
10. _____
11. _____
12. _____

Assessment

Directions: Show your complete solution on a separate sheet of paper.

A 200-N child is riding swing that is attached to ropes 3.5 m long. Find the gravitational potential energy of the child relative to the child's lowest position while riding the swing when:

a) the ropes are horizontal
b) the ropes make a 30.0° angle with the vertical
c) the child is at the bottom of the arc.

What I Need to Know

After going through this module, you are expected to:

Calculate quantities related to planetary or satellite motion. [STEM_GP12Red-IIb- 20]

What I Know

Directions: Write FACT if the statement is true and write BLUFF if the statement is false. Write your answer on the space provided.

_____1. Kepler determined the shape of each planet's orbit by triangulation from different points on Earth's orbit, using observations made at many different times of the year.

_____2. Kepler's second law implies that planet moves faster when it is farther to the Sun.

_____3. Kepler's third law implies that planets closest from the Sun orbit at a slower speed than planets closer to the Sun.

_____4. Kepler described the orbits as elliptical, not circular.

_____5. Kepler's First Law of Planetary Motion states that Planets move in ellipses with the Sun at one focus.

Lesson 5 Kepler's Law of Planetary Motion

What's New

If you were asked to draw the orbit of the earth around the sun, most likely it would be a circle with the sun at the center. However if it were true, why is it we have four seasons (Sadly, we don't feel the 4 seasons since the Philippines is at the equator) mainly spring, summer, autumn and winter.

What is It

A German Astronomer *Johannes Kepler* whose observation and analysis made being an assistant to *Tycho Brahe* enabled him in describing the motion of the planets in the solar system. He was able to establish three laws which was the ***Kepler's Laws of Planetary Motion***.

Kepler's expectation that the planets would move in perfect circles around the Sun was shattered after years of effort. He discovered the paths to be ellipses. His first law stated *"The path of each planet around the Sun is an ellipse with the Sun at one **focus**."* (Fig. 4a)

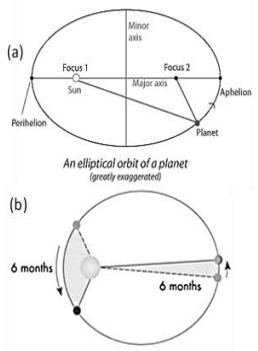


Figure 4. Kepler's Law of Planetary Motion (a) 1st Law (b) 2nd Law

Kepler also found that the planets do not revolve around the Sun at a uniform speed but move faster when they are nearer the Sun and slower when they are farther from the Sun. Imagine marking the start and end of a 6 months time period while near the perihelion and aphelion. (Fig. 4b) You will notice the that the planet spends a larger arc (not time) near the perihelion compared to the aphelion, this only accounts for the planet's speed. If you compute the areas of those time periods near the perihelion and aphelion will yield equal areas.

This is Kepler's second law, "*The line from the Sun to any planet sweeps out **equal areas** of space in **equal time** intervals.*"

What's More

Materials: Any decorative materials available at home.
Note: *Ask assistance from parent/guardian if working with potentially harmful equipment and substance.*
Objective: To create a diorama that illustrates Kepler's First Law and Second Law of Planetary Motion and 4 seasons in a year (e.g. winter, summer)
Size: Must not exceed 8.5 x 14 in

Table 4: RUBRIC FOR DIORAMA

	4	3	2	1
Accuracy	Accurately depicts source material and shows evidence of strong comprehension	Depicts source material and shows some evidence of comprehension	Unclearly depicts source material and shows little evidence of comprehension	Does not depict source material and no evidence of comprehension is shown
Creativity	Shows great creativity and thought, student created something completely new using source material	Shows great creativity and thought, student created something interesting using source material	Shows some creativity and thought	Shows little to no creativity and thought
Construction	Extremely well put-together, used very interesting materials	Well put together, some interesting materials used	Sloppy, materials are falling off and/or apart	Not construct, everything just throws without purpose
Presentation	Student spoke incredibly clearly, demonstrated great confidence and practice	Student spoke clearly, showed signs of practice	Student was unclear when speaking, showed little evidence of preparation	Student presentation made little sense, no evidence of practice or preparation

What I Can Do

Directions: Answer the following question explicitly. Answer on a separate sheet.

1. Earth is closer to the Sun in December than in June. In which of these two months is Earth moving faster around the Sun? Justify your answer.
2. At which of the indicated positions (Fig.5) does the satellite in elliptical orbit experience the greatest:
   a) gravitational force
   b) speed
   c) momentum
   d) gravitational potential energy
   e) kinetic energy
   f) acceleration
   g) total energy

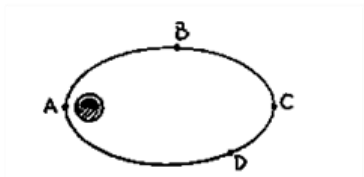


Figure 5.
Source: Hewitt, P.G. (2006). Conceptual Physics. San Francisco:Pearson Addison Wesley.

Assessment

Directions: Write the letter of the correct answer.

1. According to Kepler, the paths of planets about the Sun are:
   A. circles C. parabolas
   B. ellipses D. straight lines
2. Which of Kepler's law relates the revolving of planets around the sun in an elliptical orbit?
   A. 1st C. 3rd
   B. 2nd D. 4th
3. From Kepler's law of orbit, we can infer that the sun is located _____ of the planet's orbit.
   A. at the center C. at both foci
   B. at one of the foci D. anywhere along the semi-minor axis
4. Kepler's laws of planetary motion improved _____
   A. the heliocentric theory C. the big bang theory
   B. the geocentric theory D. the string theory
5. What is the place where a planet is farthest away from the Sun in its orbit around the Sun?
   A. foci C. aphelion
   B. axis D. perihelion

What I Need to Know

After going through this module, you are expected to:

1. **Relate the amplitude, frequency, angular frequency, period, displacement, velocity, and acceleration of oscillating systems [STEM_GP12PMIIc-24];**
2. **Recognize the necessary conditions for an object to undergo simple harmonic motion [STEM_GP12PMIIc-25]; and**
3. **Calculate the period and the frequency of spring mass, simple pendulum, and physical pendulum [STEM_GP12PMIIc-27].**

What I Know

Directions: Match the words in Column A to the units in Column B. Write the letter on the space provided. Answers may be used twice.

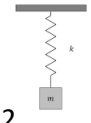
Column A	Column B
____1. amplitude	A. Hz
____2. angular frequency	B. $kg.m/s^2$
____3. frequency	C. s
____4. period	D. rad/s
____5. wavelength	E. J/s
	F. m

Lesson 6	Periodic Motion

What's New

Directions: Write *Y* if the image shows an example of motion repeated in equal intervals of time and *N* if not.

____1.

____2. 

____3.

____4.

____5.

What is It

Many kinds of motion repeat themselves over and over: the vibration of quartz crystal in a watch, the swinging pendulum of a grandfather clock, the sound vibration produced by a clarinet or an organ pipe, and the back-and-forth motion of the pistons in a car engine, the motion creates by the swing when you play on it. We call this **periodic motion** or **oscillation**. Understanding periodic motion will be essential for our later study of waves, sound, alternating electric currents and light.

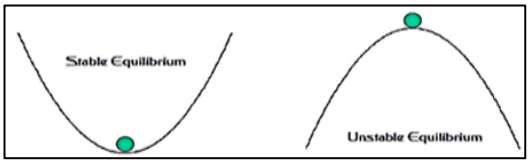


Figure 6. Stable and unstable equilibrium

Periodic Motion is a motion that repeat about a point of stable equilibrium. A necessary requirement for periodic motion is a restoring force.

CHARACTERISTICS

1. **Amplitude (A) –** this is the maximum displacement from the equilibrium position. The SI unit of amplitude is meter.
2. **Period (τ) –** this is the time for one cycle. The SI unit is the second, but it is sometimes expressed as "seconds per cycle".
3. **Frequency (*f*) –** is the number of cycles in a unit of time. The SI unit of frequency is the hertz: **1 hertz = 1 Hz = 1 cycle/s = 1 s^{-1}**
4. **Angular Frequency (ω) –** is also known as radial or circular frequency, it measures ***angular displacement*** per unit time.
5. **Wavelength** (λ) **–** is the distance between any two successive points in a wave that are in phase with each other.

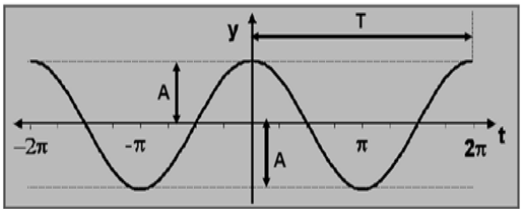


Figure 7. Characteristics of waves

What's More

Illustrative Example:

A piano string sounds a middle A by vibrating primarily at 220 Hz.

a. Calculate its period and angular frequency.

b. Calculate the period and angular frequency for a soprano singing a "high A". Two octaves up, which is four times the frequency of the piano string.

Solution:

a. $\tau = \frac{1}{f}$ = 1 / 220 Hz = 4.55 x 10^{-3} = **4.55 ms**

$\omega = 2\pi f$ = 2π(220) = **1.38 x 10^3 rad/s**

b. Since the frequency is four times thus 220 Hz times 4 is 880 Hz.

$\tau = \frac{1}{f}$ = 1 / 880 Hz = 1.14 x 10^{-3} = **1.14 ms**

$\omega = 2\pi f$ = 2π(880 Hz) = **5.53 x 10^3 rad/s**

What I Can Do

Directions: Answer the following questions on the space provided.

1. Cite an example of a system in a periodic motion and relate the following; amplitude, frequency and period. _____
_____
_____

2. Give an example of an object in simple harmonic motion and relate the different conditions of Simple Harmonic Motion. _____
_____
_____

Assessment

Directions: Solve the problem on a separate sheet of paper.

A particle is attached to a spring that creates a vibration with a frequency of 8 Hz. Determine the period of vibrations and the angular frequency.

What I Need to Know

After going through this module, you are expected to:

1. **Differentiate underdamped, overdamped, and critically damped motion [STEM_GP12PMIId-28];**
2. **Define mechanical wave, longitudinal wave, transverse wave, periodic wave, and sinusoidal wave [STEM_GP12PMIId-31]; and**
3. **From a given sinusoidal wave function infer the speed, wavelength, frequency, period, direction, and wave number [STEM_GP12PMIId-32].**

What I Know

Directions: Use the word bank to fill in the blank of the following questions. Each word will be used only once.

Transverse	Mechanical
Longitudinal	Electromagnetic
Sinusoidal	Vacuum

1. _____ waves occur when the motion of the medium is parallel to the direction of the wave.
2. _____ waves occur when the motion of the medium is at right angles (perpendicular) to the direction of the wave.

3. _____ waves that travel through a medium.
4. _____ waves that can travel through a vacuum.
5. _____ waves that show how the amplitude of a variable changes with time.

Lesson 7	Mechanical wave and Sinusoidal wave

What's New

Waves are everywhere and can be studied in terms of its forms and its method of energy transfer. But firstly, what is a wave? In simple terms, a wave is a disturbance or a vibration that transfers energy from one point to another. The transfer of energy is achieved through a medium or in some cases through vacuum (empty space). The disturbances which transfer the energy in the direction of the wave motion, travels without transferring matter. In other words, matter is not carried with the wave.

What is It

Some waves must travel through a substance known as the matter **medium**. It can be solid, liquid or gaseous. Sound, ocean and seismic waves are some waves that travel through a medium. As the wave travel through the medium, the particles of the medium vibrate perpendicular or parallel to the direction of the wave motion. Waves that travel through a medium are called **mechanical waves**.

Damped Harmonic Motion

Damped Harmonic Motion is a harmonic motion with a frictional or drag force (Fig. 8). If the damping is small, we can treat it as an "envelope" that modifies the undamped oscillation.

However, if the damping is large, it no longer resembles SHM at all (Fig. 9).

A: underdamping: there are a few small oscillations before the oscillator comes to rest.
B: critical damping: this is the fastest way to get to equilibrium.
C: overdamping: the system is slowed so much that it takes a long time to get to equilibrium.

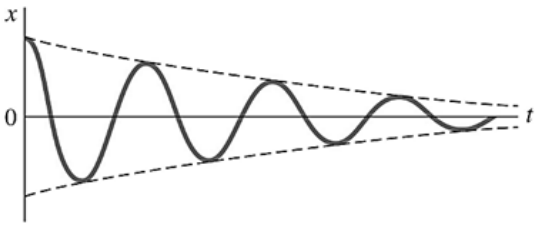


Figure 8. Damped Harmonic Motion

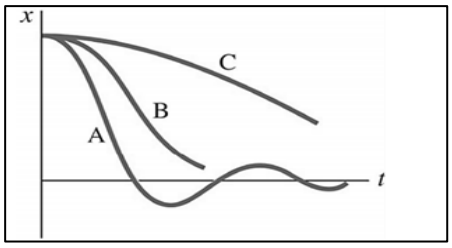


Figure 9. Kinds of Damped Harmonic Motion

Types of waves according to methods of energy transfer

When waves are classified according to their method of transfer of energy, there are two types, **mechanical** and **electromagnetic waves.**

A *mechanical wave* is a wave that is an oscillation of matter, and therefore transfers energy through a medium. Mechanical waves originate in an elastic material medium (air, steel and water). Properties which are responsible for

transmission of mechanical waves in a medium are elasticity and density of the medium.

Some examples of mechanical waves are: vibration of string, the surface wave produced on the surface of solid and liquid (water), sound waves, tsunami waves, earthquake P-waves, ultra sounds, vibrations in gas, and oscillations in spring, internal water waves, and waves in a slinky spring coil.

Electromagnetic (EM) waves are caused by varying electric and magnetic fields. As a result of the varying magnetic and electric fields a periodic change takes place creating the EM waves. Electromagnetic waves do not require any medium for its propagation and can travel through vacuum.

Two main types of wave motion

The motion of particles in a wave (Fig. 10) can either be perpendicular to the wave direction (transverse) or parallel to it (longitudinal).

(a) A traveling wave that causes the particles of the disturbed medium to move perpendicular to the wave motion is called a **transverse wave.**

(b) A traveling wave that causes the particles of the medium to move parallel to the direction of wave motion is called a **longitudinal wave.**

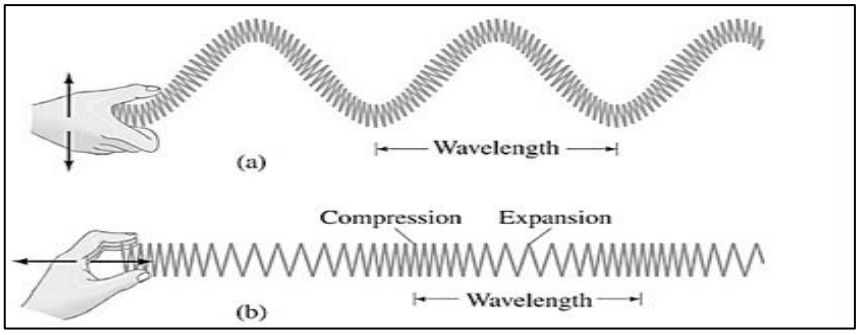


Figure 10. Types of wave motion

What's More

SINUSOIDAL WAVES

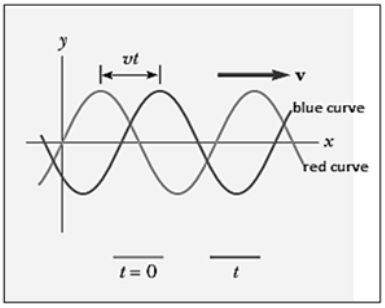


Figure 11. Sinusoidal Wave

The wave represented by the curve shown on Figure 11 is called a *sinusoidal wave* because the curve is the same as that of the function sin θ plotted against θ. The sinusoidal wave is the simplest example of a periodic continuous wave and can be used to build more complex waves. The red curve represents a snapshot of a traveling sinusoidal wave at $t = 0$ and the blue curve represents a snapshot of the wave at some later time t. At $t = 0$, the function describing the positions of the particles of the medium through which the sinusoidal wave is traveling can be written:

$$y = A\,sin\left(\frac{2\pi}{\lambda}x\right)$$

where the constant A represents the wave amplitude and the constant λ is the wavelength. Thus, we see that the position of a particle of the medium is the same whenever x is increased by an integral multiple of λ. If the wave moves to the right with a speed v, then the wave function at some later time t is:

$$y = A\,sin\left[\frac{2\pi}{\lambda}(x - vt)\right]$$

so represents a wave traveling to the right. If the wave were traveling to the left, the quantity $(x - vt)$ would be replaced by $(x + vt)$ as we learned when we developed the previous equations.

By definition, the wave travels a distance of one wavelength in one period T. Therefore, the wave speed, wavelength, and period are related by the expression

$$\boldsymbol{v = \frac{\lambda}{t}}$$

We can express the wave function in a convenient form by defining two other quantities, the angular wave number k and the angular frequency ω: $\boldsymbol{k \equiv \frac{2\pi}{\lambda}}$ **;** $\boldsymbol{\omega \equiv \frac{2\pi}{T}}$

Using these definitions, we see that the sinusoidal wave equation can be rewritten in a more compact form

$$\boldsymbol{y = A\,sin\,(kx - \omega t)}$$

(Wave function for a sinusoidal wave)

Illustrative Examples:

1. A Traveling Sinusoidal Wave

A sinusoidal wave traveling in the positive x direction has an amplitude of 15.0 cm, a wavelength of 40.0 cm, and a frequency of 8.00 Hz. The vertical displacement of the medium at $t = 0$ and $x = 0$ is also 15.0 cm, as shown in the figure on the right. Find the angular wave number k, period T, angular frequency ω, and speed v of the wave.

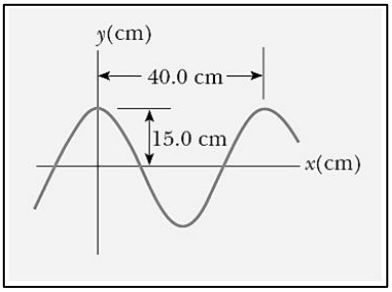


Figure 12. A Traveling Sinusoidal Wave

Solution:

$$k = \frac{2\pi}{\lambda} = \frac{2\pi\ rad}{40.0cm} = \mathbf{0.157\ rad/cm}$$

$$\omega = 2\pi f = 2\pi(8.00\ s^{-1}) = \mathbf{50.3\ rad/s}$$

$$T = \frac{1}{f} = \frac{1}{8.00\ s^{-1}} = \mathbf{0.125\ s}$$

$$v = \lambda f = (40.0\ \text{cm})(8.00\ s^{-1}) = \mathbf{320\ cm/s}$$

2. A sinusoidal wave is traveling along a rope. The oscillator that generates the wave completes 40.0 vibrations in 30.0 s. Also, a given maximum travels 425 cm along the rope in 10.0 s. What is the wavelength?

$$f = \frac{40\ vibrations}{30s} = 1.33Hz \qquad v = \frac{425cm}{10.0s} = 42.5cm/s$$

$$\lambda = \frac{v}{f} = \frac{42.5cm}{1.33Hz} = 31.9cm = \mathbf{0.319m}$$

What I Can Do

Directions: Answer the following questions on a separate sheet.

1. Differentiate transverse wave from longitudinal wave.
2. From a given sinusoidal wave function, determine the relationship among the following quantities: speed, wavelength, frequency, period, direction, and wave number.

Assessment

Directions: Solve the following problems on a separate sheet of paper.

1. For a certain transverse wave, the distance between two successive crests is 1.20 m, and eight crests pass a given point along the direction of travel every 12.0 s. Calculate the wave speed.
2. A sinusoidal wave train is described by the equation:

$$y = (0.25\text{m})sin(0.30x - 40t\ )$$

where x and y are in meters and t is in seconds. Determine for this wave the

(a) amplitude
(b) angular frequency
(c) angular wave number
(d) wavelength
(e) wave speed
(f) direction of motion

What I Need to Know

After going through this module, you are expected to:

1. **Describe qualitatively and quantitatively the superposition of waves [STEM_GP12MWSIIe-35];**
2. **Apply the condition for standing waves on a string [STEM_GP12MWSIIe-36];**

What I Know

Directions: Identify the missing word by filling in the letters to complete the sentence.

1. The D_ _ p _ _ r effect is evident when you hear the changing pitch of an ambulance siren as it passes you by.
2. Standing waves are formed by the superposition of two or more waves moving in _ _ p _ s _ t _ directions.
3. Waves superimpose on one another by adding their d _ _ t _ _ b _ _ c _ s.
4. S_ p_ _ _ o _ i _ _ _ n of waves is when two waves combine at the same place at the same time.
5. _ _ _ s _ r _ c _ _ v _ interference is the overlapping of waves exhibit which results to a greater amplitude.

Lesson 8 Superposition of waves and Standing waves

What's New

People apply physics subconsciously, by knowing to replace tires if the patterns appear faded, knowing that shouting inside an empty auditorium will produce an echo or even the placement of speakers in your living room. Why do you think the speakers in a concert are aligned (Fig. 13)? To increase the volume by overlapping the waves.

Figure 13. Speakers in a concert

What is It

In analyzing wave combination, one must remember the superposition principle, "*If two or more traveling waves are moving through a medium, the resultant value of the wave function at any point is the algebraic sum of the values of the wave functions of the individual waves.*"

Under the superposition principle is that two waves (y_1 & y_2) traveling within the same medium moving toward opposite direction will pass through each other without any change.

A wave phenomenon of having two or more waves occupying the same region of space producing a resultant wave is called **interference**. Interference has two types, *constructive interference* and *destructive interference*.

Constructive interference (Fig. 14) refers when two overlapping waves exhibit similar wave properties which will result in a greater amplitude than of those individual overlapping waves.

Destructive interference *(Fig. 15)* on the other hand will be observed if two waves interact but the displacements caused by the two waves are in opposite directions.

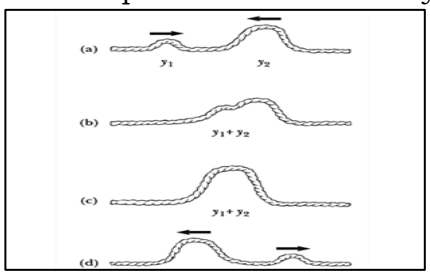


Figure 14. Constructive interference

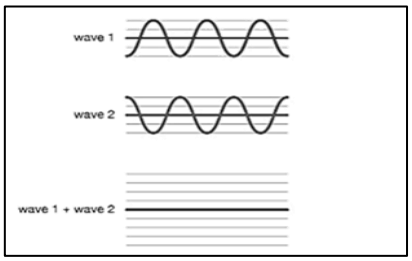


Figure 15. Destructive interference

Example 1:

Two triangular wave pulses are traveling toward each other on a stretched string as shown in the Fig. 16. Each pulse is identical to each other and travels at 2.00 cm/s. The leading edges of the pulses are 1.00 cm apart at t= 0. Sketch the shape of the string at:

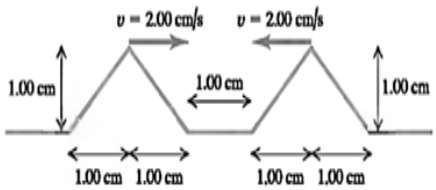


Figure 16. Two Triangular Wave Pulses

a) 0.250 s c) 0.750 s e) 1.250 s
b) 0.500 s d) 1.000 s

Solution

Let us now apply the principle of superposition to two sinusoidal waves traveling in the same direction. If the two waves are traveling to the right and have the same frequency, wavelength, and amplitude but differ in phase, we can express their individual wave functions.

$$y_1 = A\sin(kx - \omega t) \qquad y_2 = A\sin(kx - \omega t + \phi)$$

where, $k = 2\pi\lambda$, $\omega = 2\pi f$, and Φ is the phase constant, so the resultant wave function y is $y = y_1 + y_2 = A[\sin(kx - \omega t) + \sin(kx - \omega t + \phi)]$

Using a trigonometric identity to simplify this expression will result in:

$$y = 2A\cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t + \frac{\phi}{2}\right)$$

The resultant wave function y also is sinusoidal and has the same frequency and wavelength as the $\left(\frac{\phi}{2}\right)$ individual waves because the sine function incorporates the same values of k and ω that appear in the original wave functions. The amplitude of the resultant wave is $2A\cos\left(\frac{\phi}{2}\right)$ and its phase is

Example 2:

Two traveling sinusoidal waves are described by the wave functions

y_1= 5.00 m sin [π (4.00x – 1200t)] y_2= 5.00 m sin [π (4.00x – 1200t – 0.25)]

If x, y_1, & y_2 are in meters while t is in seconds. Find the amplitude and frequency of the resultant wave.

Solution

Using the equation: $y = 2A\cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t + \frac{\phi}{2}\right)$

$$y = 2A\cos\left(\frac{\emptyset}{2}\right) = 2\ (5.00)\ \cos\ (-0.25/2) = \mathbf{10.00\ m}$$

$$f = \frac{\omega}{2\pi} = \mathbf{600\ Hz}$$

Let's observe the superposition of waves in the example of wave reflections from a fixed-end or free-end boundary. Imagine a transverse wave moving in a stretched string. What happens when a wave encounters a boundary at the end of the string? If the end is fastened to a rigid support, it is a fixed-end (Fig. 17) that cannot move.

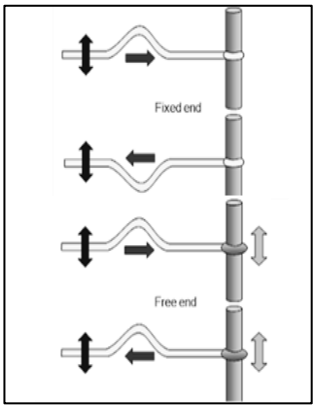


Figure 17. Fixed-end vs. Free-end

The arriving wave exerts a force on the support; the reaction to this force, exerted by the support on the string produces a reflected pulse in the opposite direction. So, when incoming waves interact with the reflected waves, destructive interference is observed. The opposite situation from an end that is held stationary is a free-end, one that is perfectly free to move in the direction perpendicular to the length of the string like being held by a person. Unlike the fixed-end boundary, this produces a reflected wave with the same direction as the incoming wave, thus increasing the amplitude and observing constructive interference. Subsequently, by producing waves with the proper condition like matching the natural frequency of the string will produce a **standing wave** (Fig. 18).

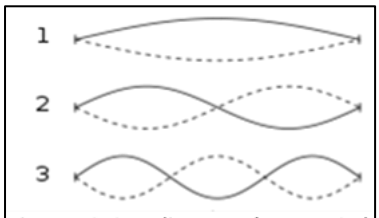


Figure 18. Standing wave's 1st to 3rd harmonic

Standing waves are set up in the string by a continuous superposition of waves incident on and reflected from the ends, the wave pattern remains in the same position along the string and its amplitude fluctuates. It is called standing wave since it doesn't appear to be moving in either direction but only seems to vibrate in place.

Standing waves do not form under just any circumstances. They require that energy be fed into a system at an appropriate frequency. That is, when the driving frequency applied to a system equals its natural frequency. This condition is known as resonance. Standing waves are always associated with resonance. Resonance can be identified by a dramatic increase in amplitude of the resultant vibrations. (Scan QR code on the right to watch a video about "Standing Wave Demo" or visit the link https://youtu.be/-k2TuJfNQ9s)

SCAN ME

Any system in which standing waves can form has numerous natural frequencies. The set of all possible standing waves are known as the **harmonics** of a system. The simplest of the harmonics is called the fundamental frequency or first *harmonic*. Increasing the frequency by a factor of 2 will produce the 2nd harmonic, and increasing the frequency by a factor of 3 will produce the 3rd harmonic.

What's More

Directions: Solve the following problems on a separate sheet of paper.

1. Two pulses are moving in opposite direction at 1.0 cm/s on a taut string. Each square is 1.0 cm. Sketch the shape of the string at the end of 6 seconds.
2. Two waves in one string are described by the wave functions

   $y_1 = 3\cos(4x - 1.6t)]$ $\qquad$ $y_2 = 4\sin(5x - 2t)$

   where y and x are in centimeters and t is in seconds. Find the superposition

of the waves at the points:

a) x = 1.00, t = 1.00 b) x = 1.00, t = 0.500 c) x = 0.500, t = 0.

What I Can Do

Directions: Use sketches/screenshots WITH REFERENCE to the video to answer the questions. It should be placed on a separate sheet.

Watch and observe the YouTube video "Overtones are awesome. Here is why". In this video, you will see a musician demonstrates the 1st harmonic along with the overtones which are observed in a frequency spectrum analyzer. Overtones are what musicians refer to those in 2nd harmonic and higher nth harmonic.

1. Sketch the 1st harmonic (without the overtones).
2. How did we perceive the 1st harmonic along with the overtones? (Label the overtones up to the 3rd overtone or the 4th harmonic)
3. Show the observed frequency for the 1st and 2nd overtones. Justify why is this true.

Assessment

Directions: Solve the problems on a separate paper.

1. Two pulses A and B are moving in opposite directions along a taut string with a speed of 2.00 cm/s. The amplitude of A is twice the amplitude of B. The pulses are shown in Figure 19 at t = 0. Sketch the shape of the string at t = 1, 1.5, 2, 2.5, and 3 s.

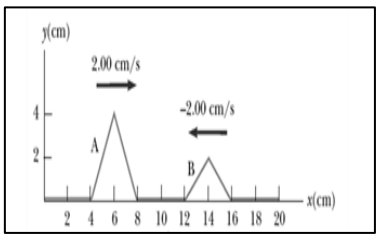


Figure 19. Two pulses in a taut string

2. Two waves in one string are described by the wave functions:

   $y_1 = 2.0\ m \sin(20.0x - 32.0\ t)$

   $y_2 = 2.00\ m \sin(25.0x - 40.0t - 0.433)$

   where y and x are in centimeters and t is in seconds. Find the superposition of the waves at the points: a) x = 1.00, t = 1.00 b) x = 1.00, t = 0.500

What I Need to Know

After going through this module, you are expected to:

Relate the frequency (source dependent) and wavelength of sound with the motion of the source and the listener [STEM_GP12MWSIIe-37].

What I Know

Directions: Write the letter of the correct answer.

1. What sort of waves exhibit the Doppler effect?
   A. light waves C. sound waves
   B. water waves D. electromagnetic waves
2. Why does the amplitude of a sound wave decrease with distance from its source?
   A. The amplitude of a sound wave decreases with distance from its source, because the frequency of the sound wave decreases.
   B. The amplitude of a sound wave decreases with distance from its source, because the speed of the sound wave decreases.

C. The amplitude of a sound wave decreases with distance from its source, because the wavelength of the sound wave increases.
D. The amplitude of a sound wave decreases with distance from its source, because the energy of the wave is spread over a larger and larger area.

3. In which of the following situations is the Doppler effect absent?
   A. The source and the observer are moving towards each other.
   B. The observer is moving toward the source.
   C. The source is moving away from the observer.
   D. Neither the source nor the observer is moving relative to one another.
4. What is the observed frequency when the observer is moving away from the source at 125m/s? The source frequency is 237 Hz and the speed of sound is 325 m/s.
   A. 303Hz C. 201Hz
   B. 259Hz D. 146Hz
5. How will your perceived frequency change if the source is moving towards you?
   A. It will be zero. C. It will become higher.
   B. It will become lower. D. It remains the same.

Lesson 9 Doppler Effect

What's New

Scan QR code on the right to watch a video of the car engine with a title "Doppler effect demonstration", starting at the 33 second mark or visit the link (https://www.youtube.com/watch?v=8RS2ToLPDZ8&t=34s). Let the students describe what they have observed by filling in the missing word of the sentence below.

SCAN ME

Doppler Effect occurs as the source of a wave _____ an observer, the observer hears a _____ frequency than the source actually is emitting. As the source _____ away from an observer, the observer hears a _____ frequency wave than the source actually is emitting.

What is It

Did you ever notice that when a motorcycle passes, its noise seems to drop as it goes farther away? This phenomenon was first described by Austrian scientist *Christian Doppler*, called the **Doppler Effect**. When a source of sound and a listener are in motion relative to each other, the *frequency* of the sound heard by the listener is not the same as the source *frequency*.

Imagine sound waves propagating from a stationary source like an ambulance ringing while at rest (Fig. 20a). The waves are evenly spaced throughout every direction; listeners will hear the same sound at the same frequency.

What if the ambulance is dashing towards you (Fig. 20b), don't you feel the frequency getting stronger? Since the distance between you and the ambulance decreased while the

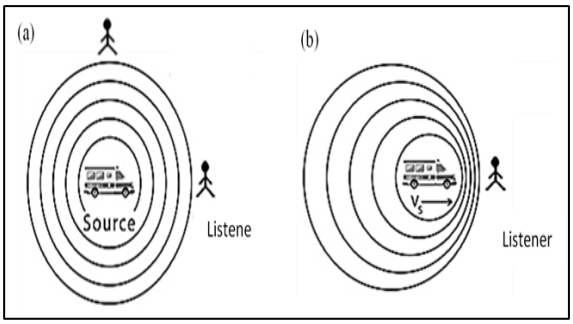


Figure 20. Doppler Effect

ambulance is still producing the same sound, which wave property changed? It's the frequency.

The Doppler effect is evident when you hear the changing pitch of an ambulance siren as it passes you. This is because the crests of the sound waves encounter your ear more frequently. And, when the vehicle passes and moves away, you hear a drop in pitch because the crests of the waves are hitting your ear less frequently.

In equation form, the *doppler effect* can be expressed as: $\boldsymbol{f_L = f_s(\frac{v \pm v_L}{v \pm v_s})}$

where:

f_L= frequency perceived by the listener
f_s= frequency perceived by the source
v= speed of sound
v_l= speed of the listener
v_s= speed of the source

This equation can change according to the direction of the source and the listener. If the listener is moving toward the source indicate a positive sign (If moving away use negative sign). If the source is moving away from the listener indicate positive sign (If towards use negative sign).

Example 1

If a listener is at rest and the siren is moving away from the listener at 30 m/s, if the siren emits a 300 Hz sound, what frequency does the listener hear?

Given

$v_L = 0$ $v_S = 30$ m/s
$f_S = 300$ Hz v = 340 m/s (speed of sound in air)

Solution

$$f_L = f_s\left(\frac{v}{v+v_s}\right) = \frac{300Hz(\,340\frac{m}{s})}{340\frac{m}{s}+\frac{30m}{s}} = \underline{\mathbf{276.68\ Hz}}$$

Example 2

The police car with its 300 Hz siren is moving toward a warehouse at 30 m/s, intending to crash through the door. What frequency does the driver of the police car hear reflected from the warehouse?

Given

$v_L = 0$ $v_S = 30$ m/s
$f_S = 300$ Hz v = 340 m/s (speed of sound in air)

Solution

This type of problem requires two steps. First, solve the frequency received by the warehouse building.

$$f_L = f_s\left(\frac{v}{v-v_s}\right) = \frac{300\ Hz\ (\,340\frac{m}{s})}{340\frac{m}{s}-30\frac{m}{s}} = 329.03\text{ Hz}$$

The second step requires us to reverse the roles of the car and the warehouse. Using that frequency of the warehouse as f_S.

$$f_L = f_s\left(\frac{v+v_L}{v}\right) = \frac{329.03\ Hz\ (340\frac{m}{s}+30\frac{m}{s})}{340\ m/s} = \underline{\mathbf{358.03\ H}}$$

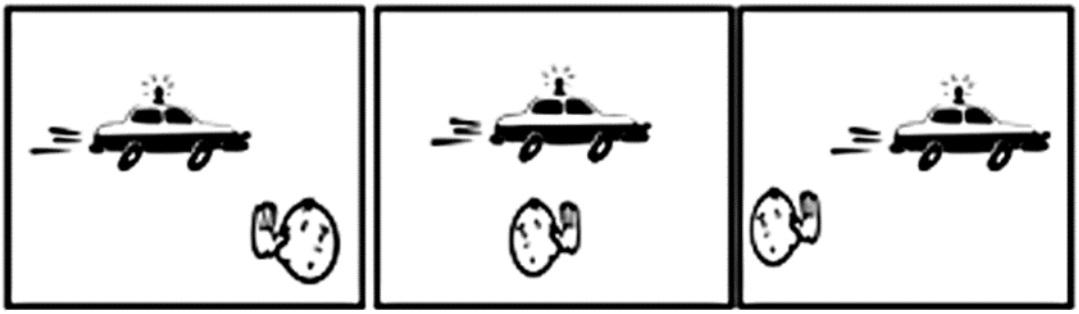
What's More

Directions: Answer and show your solution on a separate sheet of paper.

1. An unknown insect is flying towards you at 25 m/s while emitting a sound of 1200 Hz. If you remain at rest and perceive the sound at 1240 Hz, what is the speed of sound in the atmosphere?
2. The police car with its 300 Hz siren is moving away from the warehouse at 20 m/s. What frequency does the driver of the police car hear reflected from the warehouse?

What I Can Do

Directions: Write a description and explanation of what will happen to the sound at each stage of the cartoon.



1 2 3

Figure 21. Doppler Effect of a police car

_____

_____

_____

_____

Assessment

Directions: Answer the following questions explicitly on a separate sheet of paper.

1. A sound source producing 1 kHz waves moves toward a stationary listener at one-half the speed of sound. What frequency will the listener hear?
2. A car alarm is emitting sound waves of frequency of 520 Hz. You are riding a motorcycle traveling away from the car. How fast must you be traveling if you detect a frequency of 490 Hz?

What I Need to Know

After going through this module, you are expected to:

Relate density, specific gravity, mass, and volume to each other [STEM_GP12FM-iif-40].

What I Know

Directions: Write the letter of the correct answer.

1. What is the unit of specific gravity?
   A. Kg/m^3 C. m/s^2
   B. N/m3 D. $kg.m/s^2$
2. The mathematical formula to use when calculating the density of an object is:
   A. volume divided by mass C. mass plus volume
   B. mass times volume D. mass divided by volume
3. Which of the following statements is true about a bottle containing liquids that appear to be stacked?
   A. the liquid at the top is the densest
   B. the liquid at the top is the least dense

C. the liquid has the same volume
D. the liquids have the same density

4. The density of water is 1 gm/ml. An object has a mass of 58 grams. What volume must it have in order to float in water?
   A. 45 ml
   B. 50 ml
   C. 55 ml
   D. 60 ml
5. Frank has a paper clip. It has a mass of 9g and a volume of $3cm^3$. What is its density?
   A. $3\ g/cm^3$
   B. $27\ g/cm^3$
   C. $1/3\ g/cm^3$
   D. $39\ g/cm^3$

Lesson 10 Density, Specific Gravity, Mass and Volume

What's New

Directions: Read aloud the story *"Who sank the boat?"* by Pamela Allen.

Beside the sea, there once lived a cow, a donkey, a sheep, a pig, and a tiny little mouse. They were good friends, and one warm sunny morning, for no particular reason, they decided to go for a row in the bay. Do you know who sank the boat? Was it the cow who almost fell in, when she tilted the boat and made such a din? NO, it wasn't the cow who almost fell in. Do you know who sank the boat? Was it the donkey who balanced her weight? Who yelled, 'I'll get in at the bow before it's too late.' No, it wasn't the donkey who balanced her weight. Do you know who sank the boat? Was it the pig as fat as butter, who stepped in at the side and caused a great flutter? No, it wasn't the pig as fat as butter. Do you know who sank the boat? Was it the sheep who knew where to sit to level the boat so that she could knit? No, it wasn't the sheep who knew where to sit. Do you know who sank the boat? Was it the little mouse, the last to get in, who was the lightest of all? Could it be him? You DO know who sank the boat.

What is It

What is the difference between mass and volume? **Mass** is the measure of matter in an object. It is commonly used in physics; we also define it as the amount of inertia in an object. **Volume**, on the other hand, is about space, the amount of space that an object can occupy. Mass is measured in kilograms, while volume is in m^3 (We're going to use its SI Unit).

What do they have in common? Both can be utilized to measure *density*. *Density* is expressed in equation:

$\boldsymbol{\rho} = \frac{\boldsymbol{m}}{\boldsymbol{V}}$ ρ = density m = mass v = volume

An object's *density* will determine which object would sink or float. Which is denser, a person or water? A person is denser than water for most cases, except in the Dead Sea where the salinity is high which can make a person instantly float.

Comparing the density of any object to water is actually another quantity known as *Specific Gravity*, which is expressed in equation as:

$$\boldsymbol{Specific\ Gravity} = \frac{\boldsymbol{\rho_{object}}}{\boldsymbol{\rho_{water}}}$$

Specific gravity is the ratio between density, which means it has no unit. Water is known to have a density of 1000 kg/m^3. Specific gravity < 1 will make the object float on water while getting Specific Gravity >1 will make the object submerge in water.

Example

A cube is 0.38 m wide. It has a mass of 2500 g. Determine its specific gravity, and would it sink or float in water?

Given

ρ_{water} = 1000 kg/m^3 m = 2500 g or 2.50 kg V = s^3 = (0.38)^3 = 0.06 m^3

Solution

First determine the iron's density.

$$\rho_{object} = \frac{m}{V} = \frac{0.575\ kg}{0.055\ m^3} = 41.67\ \frac{kg}{m^3}$$

Second, find the specific gravity

$$Specific\ Gravity = \frac{\rho_{object}}{\rho_{water}} = \frac{41.67\ kg/m^3}{1000\ kg/m^3} = \mathbf{0.04167}$$, the object would float on water.

What's More

Activity: **Specific Gravity**

Objective: Determine the specific gravity of various household objects. Observe if they will float or sink in water.

Materials: 5 differently shaped objects

Tip: Use objects with the mass specified in the label to avoid needing a weighing scale.

Note: Ask assistance from parent/guardian if working with potentially harmful equipment and substance. Do not go outside; this is strictly done indoors.

Table 5. Solving Specific Gravity

Record your data on the space provided.				
	Name of Object	Mass(kg)	Volume(m3)	Specific Gravity
1				
2				
3				
4				
5				

What I Can Do

Directions: Answer the question on the space provided.

Would it be acceptable to replace mass with weight in computing the density of an object? Justify your answer. _____

_____

_____

Assessment

Directions: Find the angles between. Show all your pertinent solutions.

1. A cylinder with a diameter of 2 m and a height of 7 m. It has a mass of 230 g. What is its specific gravity, and would it tend to float or sink in water?
2. A rectangular prism measuring 100 cm x 200 cm x 350 cm, and has a mass of 5000 kg. What is its specific gravity, and would it tend to float or sink in water?

What I Need to Know

After going through this module, you are expected to:

1. **Relate pressure to area and force [STEM_GP12FM-IIf-41];**
2. **Relate pressure to fluid density and depth [STEM_GP12FM-IIf-42];**
3. **Apply pascal's principle in analyzing fluids in various systems [STEM_GP12FM-IIf-43]; and**
4. **Apply the concept of buoyancy and Archimedes' principle [STEM_GP12FM-IIf-41].**

What I Know

Directions: Write T if the statement is correct and F if not.

_____1. The force exerted by a fluid on the walls of its container must always by congruent to the walls.

_____2. Fluid pressure is a vector quantity.

_____3. At any particular depth, the fluid pressure is the same in all directions.

_____4. The greater the weight of the submerge object, the greater the pressure it experiences.

_____5. Pressure is force applied per unit volume.

Lesson 11 Fluid Mechanics

What's New

In this chapter, we will focus on the less rigid states of matter – liquids and gases. They are collectively known as fluids, in relation to their flowing movement. A branch of physics that is concerned with the movements of fluids is known as fluid mechanics. In definition, fluid mechanics deals with the study of all fluids under static and dynamic situations.

What is It

DENSITY AND SPECIFIC GRAVITY

- **Density** is an important property of any material and is defined as its mass per unit volume. It is denoted by the Greek letter rho (***ρ***). The SI unit of density is Kg/m^3.
- Different substances have different densities. Gases are least dense because their molecules are far apart compared to those of solids and liquids.

$$\boldsymbol{\rho} = \frac{\boldsymbol{m}}{\boldsymbol{V}}\ ;\ \frac{Kg}{m^3}\ (definiton\ of\ density)$$

where: $\rho - density, \frac{Kg}{m^3}$

$m - mass, Kg$

$V - volume, m^3$

Table 6: Mass Densities of Common Substances

Substance	Mass Density (kg/m^3)	Substance	Mass Density (kg/m^3)
SOLID		LIQUID	
Aluminum	2 700	Blood	1 060
Brass	8 470	Ethyl Alcohol	806
Copper	8 890	Glycerine	1 260
Cork	240	Mercury	13 600
Diamond	3 520	Olive Oil	910
Gold	19 300	Turpentine	870
Ice	917	Water (4 °C)	1 000
Iron	7 860	Sea Water	1 030
Lead	11 300	Air	1.29
Nickel	8 800	Carbon Dioxide	1.98
Platinum	21 500	Helium	0.178
Quartz	2 660	Hydrogen	0.0899
Silver	10 500	Nitrogen	1.25
Tin	7 290	Oxygen	1.43

SOURCE: Workbook in General Physics 1 by: Nipas, M., et. Al.; 2017

Example:

1. The body of a man whose weight is 690 N contains about $5.2x10^{-3}$ m^3 of blood. (a) Find the blood's weight and (b) express it as a percentage of the body weight.
   Solution:
   $$m = V\rho = (5.2\ x\ 10^{-3}m^3)\left(1060\frac{Kg}{m^3}\right) = 5.51\ Kg$$
   a. Blood's weight
   $$W = mg = (5.51\ Kg)\left(9.80\frac{m}{s^2}\right) = \mathbf{54\ N}$$
   b. Percentage of the body weight
   $$Percentage = \frac{54\ N}{690\ N}\ x\ 100\% = \mathbf{7.83\%}$$

2. What is the density of the liquid if a 235mL bottle has a mass of 500g when empty and 670g when filled with a liquid?
   Given:
   Volume of bottle = 235 mL
   Mass of empty bottle = 500g
   Mass of the bottle when filled with liquid = 670g
   Mass of liquid = 670g – 500g = 170g
   Solution:
   $$\rho = \frac{m}{V} = \frac{170g}{235mL} = \mathbf{0.72\frac{g}{mL}}$$

3. A cylindrical tube whose height is 2.50m and whose diameter is 4.00cm is filled with olive oil. How many kilograms of olive oil does the tube hold?
   Given:
   height (h) = 2.50m diameter (d) = 4.00cm
   radius (r) = 2.00cm or 0.02m
   Solution:
   $$V = \pi r^2 h = (3.14)(0.02)^2(2.50m) = 3.14x10^{-3}m^3$$
   $$m = \rho V = (910\frac{Kg}{m^3})\ (3.14x10^{-3}m^3) = \mathbf{2.86\ Kg}$$

What's More

PRESSURE

When a fluid is at rest, it exerts a force perpendicular to any surface in contact with it. **Pressure** is defined as the force acting perpendicular to a unit area.

$$\boldsymbol{P} = \frac{\boldsymbol{F}}{\boldsymbol{A}_{\perp}} \; ; Pascal\ (Pa)$$

where:

F is the force acting perpendicular to a given area, Newtons, N

A is the area, square meters, m^2

P is the pressure, Pascal, Pa

Example:

1. A 3500 Kg car rest on 0.9m diameter piston of a hydraulic lift in a car repair shop. How much force does the smaller piston, with a 0.4m diameter need to exert to lift the car at equal height? (Source: Villanueva, F.J.)

   Given:

   Mass of the car, m_2 = 3500Kg

   Diameter of larger piston, d_2 = 0.9m or r_2 = 0.45m

   Diameter of smaller piston, d_1 = 0.4m or r_1 = 0.2m

   Solution:

   Let F_1 *be the smaller piston*

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}; \; F_1 = \frac{F_2}{A_2}(A_1)$$

$$F_2 = m_2 g = (3\ 500Kg)\left(9.8\frac{m}{m^2}\right) = 34\ 300\ N$$

$$A_2 = \pi r^2 = (3.14)(0.45m)^2 = 0.64m^2$$

$$A_1 = \pi r^2 = (3.14)(0.2m)^2 = 0.13m^2$$

$$F_1 = \frac{F_2}{A_2}(A_1) = \frac{34\ 300\ N}{0.64m^2}(0.13m^2) = \mathbf{6967.19N}$$

BUOYANCY AND ARCHIMEDES PRINCIPLE

'Ice cubes float in a glass of water. However, pebbles sink. Why does an object sink or float in water? Archimedes, a Greek philosopher, was the first to study the ability of fluids to make things float. He stated that,' ***An object completely or partly submerged in a fluid experiences an upward force equal to the weight of the fluid displaced.'***

The buoyant force exists because fluids exert a pressure on submerged objects that increase with depth. It is an *upward force acting on an object due to fluid pressure*. It is also the weight of the fluid displaced:

$$F_B = W = mg \text{ thus, } m = \rho V \text{ therefore, } \boldsymbol{F_B = \rho V g}$$

Buoyant force results from the difference between fluid pressures on the top and bottom of an immersed solid.

$$\boldsymbol{F_B = F_2 - F_1}$$

For a body to float on liquids, such as seawater, that body must be less dense than the liquid. Ships, boats and submarines are made of materials denser than seawater yet they do not sink because they are built in such a way that a large part of their surface area is in contact with the water. However, balloons are filled with helium gas which has a density lower than air. This makes balloon float.

Example:

1. An anchor of mass 100 Kg and volume 4.0 x $10^{-2}m^3$ is resting on the bottom of a lake with a rope of negligible mass attached to it. What force must be

exerted on the rope to lift the anchor when it is: totally submerged; halfway out of the water and completely above the water?
Solution:

$$F_{anchor} = m_{anchor}g = (100Kg)\left(9.8\frac{m}{s^2}\right) = 980\ N = F_{anchor}$$

a. Totally submerged

$$F_B = \rho gV = \left(1030\frac{Kg}{m^3}\right)\left(9.8\frac{m}{s^2}\right)(4.0x10^{-2}m^3)$$
$$\boldsymbol{F_B = 403.76\ N}$$

b. Halfway of the water
Hence the volume of the water it displaces is half of the given volume $4.0x10^{-2}m^3$, or $2.0x10^{-2}m^3$

$$F_B = \rho gV = \left(1030\frac{Kg}{m^3}\right)\left(9.8\frac{m}{s^2}\right)(2.0x10^{-2}m^3)$$
$$\boldsymbol{F_B = 201.88\ N}$$

c. Completely above the water
There is no buoyant force when the anchor is above water.

What I Can Do

Directions: Answer the question on the space provided.
Why is mercury used instead of water as the liquid in the construction of a barometer? _____
_____

Assessment

Directions: Analyze and solve the following problems and show your complete solution on a separate sheet of paper.

1. What is the volume of an iceberg whose mass is 3.6 x 10^5 Kg?
2. A nurse pushes the plunger of a hypodermic syringe with a force of 4.0N. If the internal radius of the cylinder is 0.80cm and the radius of the core of the needle is 0.20mm, with what force is the vaccine ejected from the needle?
3. A crown that is supposed to be pure gold has a mass of 850g. Its apparent mass when immersed in water is 798g.
   a. What is the volume of the crown?
   b. Is it pure gold?

What I Need to Know

After going through this module, you are expected to:

1. **Explain the connection between the zeroth law of thermodynamics, temperature, thermal equilibrium, and temperature scales [STEM_GP12TH-IIg-49];**
2. **Convert temperatures and temperature differences in the following scales: Fahrenheit, Celsius, Kelvin [STEM_GP12TH-IIg-50];**
3. **Define coefficient of thermal expansion and coefficient of volume expansion [STEM_GP12TH-IIg-51];**
4. **Calculate volume or length changes of solids due to changes in temperature [STEM_GP12TH-IIg-52].**

What I Know

Directions: Write TRUE if the statement is correct. Otherwise, write FALSE.

_____1. Liquids expand on heating.
_____2. The temperature is directly related to the thermal expansion.
_____3. Most materials will expand when their temperature increases due to the motions of the microscopic particles they are composed.
_____4. The increase on length due to thermal expansion depends on the properties of the material.
_____5. The length of a material decreases due to increase in temperature.

Lesson 12 Heat and Thermodynamics

What's New

Thermodynamics is the branch of physics that deals with the relationships between heat and other forms of energy. In particular, it describes how thermal energy is converted to and from other forms of energy and how it affects matter. For instance, have you ever wondered why metal feels colder to touch than a wood? or have you ever wondered why water heats up or cools down slower than many other common substances? These are just some of the questions that can be explained using the concepts in thermodynamics.

What is It

What is Zeroth Law of Thermodynamics?

What will happen if two objects, one with higher temperature and one with lower temperature, are brought in contact with each other?

The zeroth law of thermodynamics says that if two objects are in thermal equilibrium with a third, then they are in thermal equilibrium with each other. Then you can say that each of these objects has a thermal property that they all share — this property is called **temperature**. When a body 'A' is in thermal equilibrium with another body 'B', and also separately in thermal equilibrium with a body 'C', then body 'B' and 'C' will also be in thermal equilibrium with each other.

Temperature Scales

There are three temperature scales in use today, Fahrenheit, Celsius and Kelvin.

1. **Fahrenheit Scale** - scale based on 32° for the freezing point of water and 212° for the boiling point of water, the interval between the two being divided into 180 equal parts.
2. **Celsius Scale** - the scale of temperature in which water freezes at 0° and boils at 100° under standard conditions.
3. **Kelvin Scale** - a scale of temperature with absolute zero as zero, and the triple point of water as exactly 273.16 degrees.

Celsius to Fahrenheit	° F = 9/5 (° C) + 32
Kelvin to Fahrenheit	° F = 9/5 (K - 273) + 32
Fahrenheit to Celsius	° C = 5/9 (° F - 32)
Celsius to Kelvin	K = ° C + 273
Kelvin to Celsius	° C = K - 273
Fahrenheit to Kelvin	K = 5/9 (° F - 32) + 273

Table 7. Temperature Conversion

Thermal Expansion

The expansion of alcohol in a thermometer is one of many commonly encountered examples of thermal expansion, which is the change in size or volume of a given system as its temperature changes. The most visible example is the expansion of hot air. When air is heated, it expands and becomes less dense than the surrounding air, which then exerts an (upward) force on the hot air and makes steam and smoke rise, hot air balloons float, and so forth. The same behavior happens in all liquids and gases, driving natural heat transfer upward in homes, oceans, and weather systems, as we will discuss in an upcoming section. Solids also undergo thermal expansion. Railroad tracks and bridges, for example, have expansion joints (Fig. 22) to allow them to freely expand and contract with temperature changes.

Figure 22. Road bridges and expansion joints

Thermal expansion is the property that causes metals to expand in size as the temperature goes up, and to shrink in size as the temperature goes down.

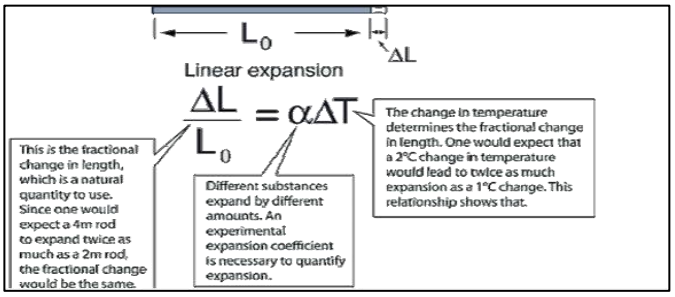


Figure 23. Linear Expansion

Heat and Heat Capacity

Heat is the transfer of energy from the higher temperature object to a lower temperature object. The heat capacity measures the amount of heat necessary to raise the temperature of an object or system by one degree Celsius.

Specific heat capacity is the amount of heat energy required to raise the temperature of a substance per unit of mass. The specific heat capacity of a material is a physical property. It is also an example of an extensive property since its value is proportional to the size of the system being examined.

In SI units, specific heat capacity (symbol: c) is the amount of heat in joules required to raise 1 gram of a substance 1 Kelvin. It may also be expressed as J/kg·K. Specific heat capacity may be reported in the units of calories per gram degree Celsius, too.

$$Q = m\,c\,\Delta T$$

Q = energy transfered (joules)
m = mass of water (grams)
c = specific heat capacity
ΔT = temperature change (K or ^{o}C)

What's More

Activity 1: Zeroth Law of Thermodynamics Experiment

Materials:

100 mL tap water	large bowl
thermometer	100 mL cold water from the ref

Procedures:

1. Using the thermometer, get the temperature of the tap water and the cold water, respectively. Take note of the temperature readings in degree Celsius.
2. In a large bowl, pour the tap and the cold water. Leave for about 5 minutes.
3. Using the thermometer, record the temperature of the solution.

Guide Questions:

1. What are the temperature readings of the tap and cold water, respectively?

_____

2. What is the temperature reading of the solution of tap and cold water in a large bowl? Is it higher or lower than the initial temperature readings of tap and cold water? Why?

_____

_____

Activity 2: **Specific Heat Capacity Experiment**

Materials:

2 rubber balloons	candle
water	match/lighter

Procedures:

1. Inflate the first rubber balloon. Fill up the second balloon with about 100mL of water.
2. Lit the candle using the match/lighter.
3. Position the first balloon above the flame. Do the same for the second balloon.

Guide Questions:

1. What happened to the first balloon when placed over the fire? Why?

   _____

2. What happened to the second balloon when placed over the fire? Why?

   _____

What I Can Do

Directions: Answer what is asked on each item. Write your answers/solutions on a separate sheet of paper.

1. Complete the table using the conversion table of temperature.

Celsius	Fahrenheit	Kelvin
79		
		356
	117	
37		
	795	

2. What do you think is the reason why bridges and roads have joints? Explain.

Assessment

Directions: Essay Writing. Answer the question in one paragraph of not less than five (5) sentences.

Three substances are added to a mug to make coffee: the coffee, which is 65°C, the milk, which is 65°C, and the sugar, which is in thermal equilibrium with the coffee. Describe the thermal state of the sugar. *Note: Be guided by the following rubric.*

_____

_____

_____

_____

Areas of Assessment	5 points Excellent	4 points Very Good	3 points Good	1 point Poor
Ideas and understanding	Presents ideas in original manner and shows strong understanding	Presents ideas in consistent manner and shows clear understanding	Ideas are too general but shows adequate understanding	Ideas are vague or unclear and shows little understanding
Quality and Organization	Very informative and well-organized	Somewhat informative and organized	Gives some new information and a bit organized	Gives no new information and poorly organized
Grammar Usage and Mechanics	Virtually no spelling, punctuation or grammatical errors	Few spelling and punctuation errors, and minor grammatical errors	A number of spellings, punctuation or grammatical errors	Many spelling, punctuation and grammatical errors that interfere with the meaning

What I Need to Know

After going through this module, you are expected to:

1. **Enumerate the properties of an ideal gas. [STEM_GP12GLTIIh-57]**
2. **Solve problems involving ideal gas equations in contexts such as, but not limited to, the design of metal containers for compressed gases. [STEM_GP12GLTIIh-58]**
3. **Interpret PV diagrams of a thermodynamic process. [STEM_GP12GLTIIh-60]**
4. **Differentiate the following thermodynamic processes and show them on a PV diagram: isochoric, isobaric, isothermal, adiabatic, and cyclic. [STEM_GP12GLTIIh-63]**

What I Know

Directions: Write TRUE if the statement is correct. Otherwise, write FALSE.

_____1. According to Boyle's law, the volume of a gas is inversely proportional to its pressure at constant temperature.

_____2. According to Gay-Lussac's law, pressure is directly proportional to temperature at constant volume.

_____3. At constant temperature and a fixed amount of gas, when the pressure changes from P to 2P, the volume doubles.

_____4. At constant pressure and a fixed amount of gas, when the temperature increases from 20°C to 40°C, the volume doubles.

_____5. At constant pressure and constant volume, when the temperature changes from 100°C to 200°C, the number of gas molecules decreases by 50%.

Lesson 13 Ideal Gas Law

What's New

Understanding the behavior of molecular motion (molecular arrangement) in the three state or phase. Molecules are represented by the dots The illustration gives us an idea how molecules move and behave in this phase or state.

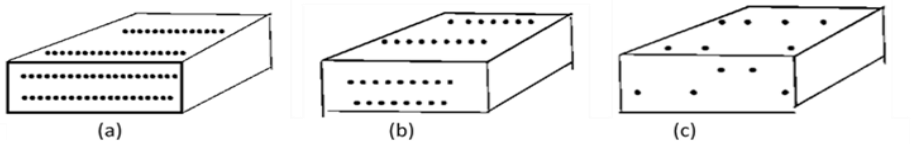


Figure 24. States of Matter

In *solid state*, molecules are completely arranged closely with one another. In *liquid state*, molecules move in any direction not as close to one another and when these molecules vibrate, some forces between are offset. In *gaseous state*, molecules are very far from one another. The forces between molecules have no effect and they move rapidly far apart.

What is It

Ideal Gas Law

Ideal gas is a theoretical gas and non - ideal gas is the real gas like oxygen. It consists of gas particles of very tiny molecules inside a cylinder (shown on Fig. 25). Gas molecules in the vessel exerts pressure on the wall that increase the pressure. Ideal gas law formula is the combination of the different gas law. **PV = nRT**

Figure 25. A cylinder with gas molecules

where:

P = pressure of the gas, atmosphere, mm of Hg

V = volume or space occupied by the gas, liters

n = number of moles of gas, moles

R = molecular gas constant = 0.0821 L. atm/K. mole

T = temperature, Kelvin

Properties of an Ideal Gas

1. The ideal gas follows certain gas laws relating the pressure, volume and temperature of gas.
2. The volume occupied by the gas molecules is very small and can be negligible.
3. In any condition, the molecular force of attraction of molecules is also negligible.

Below are some of the gas laws related to the ideal gas property that can be measured.

a. Boyle's law

- Shows the relationship between volume and pressure of the gas.
- States that at constant temperature, the volume of the gas is inversely proportional to the gas pressure.

  Then, PV = T or $P_1V_1 = P_2V_2$

  (T = constant, **isothermal** process)

Figure 26. PV diagram of

b. Charles' law

- Shows the effect of temperature on the volume of gas.

- States that at constant pressure, the volume of the gas is directly proportional to the absolute temperature so, $\frac{V}{T} = P$ or $\frac{v_1}{T_1} = \frac{V_2}{T_2}$

  (P = constant, **isobaric** process)

c. Gay Lussac's law

- Shows the effect of temperature on the pressure of gas.
- States that at constant volume, the pressure of the gas is directly proportional to the absolute temperature so, $\frac{P}{T} = V$ or $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

  (V = constant, **isometric** process)

These gas laws were combined which resulted to a single equation called the **ideal gas law**. It also shows the relationship between pressure, volume and temperature when ideal gas changes from different state or phase. The relationship can be drawn in a P, V, T diagram and can be expressed mathematically **PV = nRT,** where *n* is the number of moles and R = 8.314 J/mole.K.

What's More

Illustrative Example:

1. A 60-liter metal cylinder contains 45 number of moles of Hydrogen gas. Find pressure inside the metal cylinder when gas temperature is 298 K?

   Given:

   Gas = hydrogen V = 60 liters and T = 298 K n = 45 moles

   R = 0.0821 L.atm /K.mole (constant)

   Find: P (units are compatible)

$$\mathbf{P = nRT/V}$$

$$\mathbf{P} = \frac{nRT}{V} = \frac{(45\ mole)(\ 0.0821\ L..atm\ /Kmole)(\ 298\ K\ )}{60\ liters} = \mathbf{18.35\ atm}$$

What I Can Do

Directions: Solve the problem using the different steps.

1. The initial pressure of the gas in an enclosed cylindrical tank is 75KPa at 305K. What is the pressure of the gas when the temperature becomes 398K?

Assessment

Directions: Write the letter of the correct answer before each number.

1. Which of the following terms is NOT part of ideal gas law?
   A. Time C. Pressure
   B. Volume D. Temperature
2. What is the shape of a P-T curve for ideal gas?
   A. Ellipse C. Hyperbolic
   B. Parabolic D. Straight Line
3. Which law shows the relationship between pressure and volume of gas?
   A. Boyle's Law C. Ideal Gas Law
   B. Charles' Law D. Avogadro's Law
4. What is the shape of a P-V curve for an ideal gas?
   A. Ellipse C. Hyperbolic
   B. Parabolic D. Straight Line
5. What is the pressure of a 5 liter/mole ideal gas at temperature 27°C?
   A. 2.5 atm C. 4.4 atm
   B. 3.8 atm D. 5.3 atm

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For inquiries or feedback, please write or call:

Department of Education – SDO Valenzuela
Office Address: Pio Valenzuela Street, Marulas, Valenzuela City
Telefax: (02) 8292-4340
Email Address: sdovalenzuela@deped.gov.ph